

Jasivan Alex SIVAKUMAR

Cambridge, UK (Open to global relocation)

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RESEARCH PROFILE

NLP Researcher with **PhD** focusing on numerical reasoning in large language models, alongside recent contributions to formal mathematical reasoning, benchmark evaluation, and multilingual post-training. Combines top-tier publication record (**ICLR**, **ACL**) with industry research experience at **Amazon** Alexa and **Huawei** Noah's Ark Lab.

PROFESSIONAL EXPERIENCE

- Sep 25 - Dec 25
(Bellevue, US) **Amazon - Alexa International**, Applied Scientist Intern
- Developed multilingual reward models for DPO and Bradley–Terry post-training across seven languages and six cultural styles, revealing a systematic 10% performance gap in frontier LLMs, and found that humans identify 12% more preference ties than LLMs.
- Apr 25 - Sep 25
(London, UK) **Huawei - Noah's Ark Lab**, Research Scientist Intern
- Refactored Lean 4 formalisation to build a 456 Natural Language–Lean dataset with occulted answers; presented insights to senior leadership, resulting in a lead-author paper and a second co-authorship via cross-team collaboration.
- Jul 24 - Oct 24
(Bellevue, US) **Amazon - Alexa International**, Applied Scientist Intern
- Architected and shipped a human-in-the-loop evaluation protocol for multi-model benchmarking across any language using AWS; built a core production pipeline that remains actively utilised since deployment.
- Jan 24 - Jun 26
(Sheffield, UK) **Speech and Language Technology Hub**, Consultant
- Advised and proposed AI solutions to four clients: RAG for automated query response, speech processing for stress detection, topic modeling for document analysis, and cross-lingual token mappings; driving 75% client progression to active implementation.

SELECTED PUBLICATIONS

- Oct 26
(INLG) **Sivakumar, J., & Moosavi, N. (2026).** [Decoding Path Recovery in Number Generation](#) International Conference on Natural Language Generation
- Analyse LLM numerical error recoverability across 8 language models, revealing that 30% of arithmetic failures are recoverable from lower rank predictions or intermediate layers, identify probability/rank dynamics that signal potential incorrect token selection.
- Apr 26
(ICLR) Paul, D., et. al. including **Sivakumar, J. (2026).** [A Benchmark for Deep Information Synthesis](#) International Conference on Learning Representation.
- Authored a multi-step benchmark problem with complete data retrieval steps, Python analysis code, and structure outputs that remain unsolved across 12 Agentic AI models.
- Oct 25
(preprint) **Sivakumar, J., Borchert, P., et. al. (2025).** [Conjecturing: An Overlooked Step in Formal Mathematical Reasoning](#) ArXiv
- Constructed ConjectureBench to reveal a 24% accuracy drop in LLMs when answers are occulted, and engineered a decoupled Lean-CoT pipeline with few-shot learning to boost end-to-end autoformalisation and verified proof performance by up to 18.4%.
- Jan 25
(COLING) **Sivakumar, J., & Moosavi, N. (2025).** [How to Leverage Digit Embeddings to Represent Numbers?](#) International Conference on Computational Linguistics.
- Introduced mathematically grounded digit embedding aggregations via input tokens and auxiliary loss functions, achieving up to 6.17% accuracy gain in numerical reasoning without needing any architectural modifications.
- Jul 23
(ACL) **Sivakumar, J., & Moosavi, N. (2023).** [FERMAT: An Alternative to Accuracy for Numerical Reasoning](#). Association for Computational Linguistics.
- Developed a multi-view diagnostic benchmark revealing that LLMs drop up to 20% accuracy under number distribution shifts, but gain 41% with diverse templates fine-tuning, and exposed context memorisation as 63% of operations seen in training failed at test time.

EDUCATION

Sep 21 - Sep 26	PhD Computer Science - University of Sheffield, UK ▷ Thesis: Numerical Reasoning for General-Purpose Language Models
Sep 20 - Sep 21	MA Computational Linguistics (Distinction) - University of Wolverhampton, UK ▷ Thesis: Semantic Tree Biased Sentence Embeddings for NLI using Graph Neural Network
Sep 16 - Aug 17	Post-Graduate Certificate in Education - University of Cambridge, UK ▷ Thesis: How Scaffolding Can Help to Build Confidence in Compound Measures for 9W4.
Oct 12 - Jul 16	Bachelor (MMath) Mathematics (First Class) - University of Warwick, UK ▷ Thesis: Stochastic & Differential Equation Modeling of English-French Lexical Adoption

SKILLS

Research Expertise:	Pre/Post-training · Fine-Tuning · Multilingual NLP · Numerical Reasoning · Formal Reasoning · Benchmark Evaluation · Data Generation · Prompting · RAG
Core Stack:	Python · Lean4 · Bash · HPC · Unix/Linux · PyTorch · Hugging Face · scikit-learn
Languages:	French, English, Tamil (Native) · Spanish (Fluent) · German (Conversational)

RESEARCH & ACADEMIC ACTIVITIES

Research Projects	
Sep 21 - Sep 26 (Sheffield, UK)	NLP for Endangered Language Revitalisation in Colombia ▷ Secured a £1,000 fieldwork grant to collect primary data in Colombia; digitised text via OCR and supervised undergraduate developers to build a game-based learning platform.
Nov 21 - Jun 22 (Sheffield, UK)	Speech Processing and NLP to improve Video Search functionality ▷ Evaluated five NER systems across different fuzzy-matching thresholds to mitigate ASR noise; optimal setup doubled NER retrieval to 56.3%, beating Google Speech API by 11.9%.
Invited Talks & Lectures	
May 26, Oct 25	Beneath the surface of LLMs - Universidad de Rio do Sul, Università di Napoli L'Orientale
Nov 25, Oct 25	Conjecturing - Amazon Formal Language Seminar, Sheffield NLP Seminar
Feb 25	Chatbot for Legal Advice - Sheffield Law CS Seminar
Dec 24	Numerical Reasoning - Huawei Noah's Ark Lab (NLP)
Jun 23	FERMAT - Sheffield 3rd Speech and Language Technology Conference
Teaching & Service	
Jan 24 - Present	Peer Reviewer- ARR, ACL, EMNLP, NAACL, LREC, COLING, ICLR, ICML
Nov 23 - Jun 24	MSc Supervisor & Reading Group Organiser - NLP Group, University of Sheffield
	Teaching Assistant
Jan 22 - Feb 23	▷ Text Processing, Professional Issues, Mathematics for CS (University of Sheffield)
Jan 21 - May 21	▷ Mathematics of Machine Learning (University of Wolverhampton)

ADDITIONAL PUBLICATION

May 26 (PoliticalNLP - LREC)	Pastorino, V., Sivakumar, J. , & Moosavi, N. (2026). Decoding News Narratives: A Critical Analysis of Large Language Models in Framing Detection Language Resources and Evaluation Conference ▷ Evaluated FLAN-T5 and GPT architectures on news framing detection; demonstrated that fine-tuned FLAN-T5 achieves 78.83% in-domain accuracy (outperforming GPT-4 by 11.9%), but drops by 10% under out-of-domain shift; revealed models' tendencies to conflate emotional tone with structural framing.
Dec 21 (IALP IEEE)	Sivakumar, J. , et. al. (2021). A GRU-based pipeline approach for word-sentence segmentation and punctuation restoration in English . International Conference on Asian Language Processing, Institute of Electrical and Electronics Engineers. ▷ Developed a PyTorch GRU pipeline for joint whitespace and punctuation restoration, proving that refining retrieval order yields a 72% edit-distance reduction, significantly improving readability.